

Dear Editor,

We would like to submit our manuscript, “Adversarial Vulnerability of Machine-Learning GNSS Spoofing Detectors to Physically Realizable Attacks,” for consideration as an Original Article in *Satellite Navigation*.

Why this manuscript belongs in *Satellite Navigation*

Machine-learning spoofing detectors are moving from research prototypes toward integration in receivers, but their behaviour under a deliberate, adaptive adversary has not been established the way their clean-data accuracy has. Our manuscript reports a model-fair adversarial evaluation of fourteen detectors, spanning classical and deep architectures, on receiver observables generated from the Texas Spoofing Test Battery by an open software-defined receiver. Every detector is placed at a common operating point and attacked with one shared suite, including a gradient-free decision-based attack that gives non-differentiable detectors a fair test and a signal-validity enforcer that keeps every reported attack physically realizable. The central finding is that architecture choice trades detection utility against robustness rather than providing a security guarantee: no family is robust under a matched attack, and the apparent robustness some detectors show under transfer attacks is gradient masking rather than a real property. This speaks directly to resilient PNT, a core scope of the journal, and to the growing literature on learning-based GNSS spoofing detection that the journal has published.

Policy issues

We are not aware of any policy issues relevant to this submission.

Competing interests

The authors declare that they have no competing interests.

Prior publication and authorship

This manuscript has not been published and is not under consideration for publication elsewhere, in whole or in part. All authors have seen and approved the manuscript in its current form and agree to its submission to *Satellite Navigation*.

Special issue

This submission is not intended for a special issue.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

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On behalf of all authors

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